

Date: 2026-07-13

Device: Rishabh Khare's iPhone — iPhone 17

iOS: 26.5.1

App: com.2ergoTOI.jayant

Automation: Appium 3.x + XCUITest

EXECUTIVE SUMMARY

This report covers the automated QA test cycle for **TOI News iOS v18.0.1** executed on a physical **iPhone 17** running **iOS 26.5.1**. Three test categories were executed: App Cold Start Performance, Monkey Stress Test (20 minutes), and Rail Items UTM Parameter Analysis.

The app demonstrated **zero crashes** across all test runs and excellent warm-restart performance (~850 ms). True cold start times (7.7–9.6 s) exceed the 5,000 ms industry threshold and require profiling attention. All three rail items audited are missing standard UTM attribution parameters, representing a missed analytics opportunity. All reported ANR events are automation-infrastructure (XCTest WDA) latency — not app-level hangs.

Overall Status: **APPROVED WITH ACTION ITEMS**

4/5

COLD START RUNS PASSED

1 Exceeded Threshold

0

APP CRASHES (20-MIN STRESS)

PASS ✓

7,755 ms

TRUE COLD START TIME

Exceeds 5,000 ms

0/3

UTM RAIL ITEMS PASSING

All Missing UTM Params

TEST RESULTS SUMMARY

TEST #	TEST NAME	RESULT	KEY METRIC
01	App Start Time Performance (Cold Start)	CONDITIONAL PASS	True Cold: 7,755 ms Warm Avg: 862 ms
02	Monkey Stress Test (20 Minutes)	PASS	0 Crashes · 0 App-Level ANRs · 6 Screens
03	Rail Items — UTM Parameter Analysis	FAIL	0 / 3 items have required UTM parameters
04	Push Notification Performance	PASS	5 / 5 scenarios passed (100%)

ITERATION RESULTS

Iteration	Cold Start Time (ms)	Start Type	Result	Delta vs Threshold
1	7,755 ms	True Cold Start (binary read from flash)	FAIL	+2,755 ms over limit
2	4,092 ms	Partial Warm (pages partially cached)	PASS	-908 ms under limit
3	735 ms	Warm Restart (full OS page cache hit)	PASS	-4,265 ms under limit
4	4,285 ms	Partial Warm (pages partially cached)	PASS	-715 ms under limit
5	850 ms	Warm Restart (full OS page cache hit)	PASS	-4,150 ms under limit
Average	3,543 ms	—	4 / 5 PASS	—

Methodology: terminateApp → 2.5 s settle → activateApp → poll for XCUIElementTypeTabBar visible. Alert threshold: 5,000 ms (macOS osascript notification on breach).

CROSS-SESSION COLD START COMPARISON

Session	Time	Run 1 (True Cold)	Warm Avg (Runs 2-5)	Notes
Session 1	13:32	9,593 ms FAIL	840 ms	Baseline — full flash I/O cold start
Session 2	14:32	7,755 ms FAIL	2,493 ms*	Slight device warm-up vs Session 1
Session 3	14:37	904 ms (WARM)	862 ms	All warm — binary cached from Session 2

*Session 2 warm avg includes two lukewarm runs (4,092 ms, 4,285 ms) where OS had partially evicted binary pages. Session 3 (5/5 PASS, avg 862 ms, $\sigma = 25$ ms) confirms excellent warm-restart performance.

FINDING — TRUE COLD START EXCEEDS THRESHOLD

- True cold start is consistently **7.7–9.6 s** across two independent sessions, exceeding the 5,000 ms industry benchmark by +2.7–4.6 s.
- Warm restarts are excellent at **~840–900 ms** — well within threshold.
- Root cause likely:** Heavy SDK initialisation in `applicationDidFinishLaunching` (analytics, ads, push, crash-reporting SDKs) blocking the main thread before first render.

RECOMMENDATION

- Profile `applicationDidFinishLaunching` and `application(_:didFinishLaunchingWithOptions:)` using Instruments → App Launch template.
- Defer non-critical SDK initialisation (analytics, ads) to after first frame render using `DispatchQueue.main.asyncAfter(deadline: .now() + 0.5)` or a background queue.
- Target: reduce true cold start to < 5,000 ms. Re-test after each SDK deferral to measure incremental gain.

KEY METRICS

Metric	Value	Result
Test Duration	22.9 minutes	—
Total Events Processed	33 events (avg 1.4 events/min)	—
App Crashes	0	PASS
ANR / Freeze Events	8 (all WDA infrastructure — not app-level)	INFRA ONLY
App-Level ANRs	0	PASS
Overall App Stability	TOI app remained running & responsive throughout	EXCELLENT

6 UNIQUE SCREENS ACCESSED

#	Screen / Activity Name	Type	First Seen	Events
1	Home Feed	Main App	127 s	1
2	Article / Feed (Unknown Screen)	Content	184 s	4
3	Settings	Settings	372 s	1
4	App Exclusives	Main App	683 s	1
5	Article View	Content	701 s	2
6	ePaper	Content	1080 s	1

Screens targeted but not confirmed within time budget: Side Navigation, Search, Login, OTP, Category/Section, Markets, Explore. Navigation was attempted but WDA touch injection latency (15–30 s per tap after each app reset) exhausted the available time window before screen predicates could be confirmed.

Methodology: Appium 3.x + XCUITest Driver 11.1.6 via iproxy USB tunnel. Phase 1 (0–18 min): systematic screen tour of 12 targets. Phase 2 (18–20 min): pure random monkey (38% tap · 25% swipe↑ · 13% swipe↓ · 10% back). ANR threshold: 8 s. Crash detection: `mobile: queryAppState`.

MULTI-RUN SUMMARY

Run	Time	Duration	Events	Screens	Crashes	ANRs (Infra)
Run 1	21:01	20.1 min	37	6	0	15
Run 2	22:12	~10 min*	—	5	0	7
Run 3 (Final)	22:27	22.9 min	33	6	0	8

*Run 2 was terminated at 10 min to apply WDA warm-up fix. Consistent finding across all runs: **0 app crashes**. All ANRs are XCTest/WDA infrastructure (iOS 26.5.1 touch injection re-attach latency), not app-level.

ROOT CAUSE — IOS 26.5.1 XCTEST TOUCH INJECTION LATENCY

- After every `terminateApp + activateApp` reset, XCTest's pointer-event injection pipeline re-attaches to the newly spawned app process. On iOS 26.5.1, this re-attachment takes **15–30 s per touch event** until the pipeline is established; subsequent taps settle to < 600 ms.
- With 8 resets across 20 minutes, each reset burns 30–60 s of warm-up budget, limiting the number of unique screens confirmable within the time window.
- **This is an iOS 26 XCTest framework constraint, not a TOI app issue.**

RECOMMENDATION — AUTOMATION IMPROVEMENT

- Replace `terminateApp + activateApp` resets with UI-level navigation back to the home tab bar tap. This avoids breaking the XCTest event pipeline and maintains tap latency at < 600 ms throughout the test.
- Expected outcome: 20+ unique screens confirmable within the same 20-minute window.

OVERALL RESULT: ✗ 0 / 3 ITEMS PASS

UTM keys checked: `utm_source` · `utm_medium` · `utm_campaign` · `utm_content` — All four required for a PASS.

RAIL ITEM	DESTINATION	BROWSER	UTM PARAMS	RESULT
AI Masterclass	timesofindia.indiatimes.com/toi/ai-masterclass	WKWebView	None (has <code>ag=toiapprails</code> only)	FAIL
Chat with Astrologer	api.whatsapp.com (domain only)	SFSafariViewController	None (URL restricted by iOS)	FAIL
Financial Freedom	economictimes.indiatimes.com (domain only)	SFSafariViewController	None (URL restricted by iOS)	FAIL

ITEM-BY-ITEM BREAKDOWN

RAIL ITEM	DESTINATION	BROWSER	PARAMS FOUND	ALL 4 UTM KEYS
AI Masterclass FAIL	timesofindia.indiatimes.com/toi/ai-masterclass	WKWebView	<code>ag=toiapprails</code> only	✗ All Missing
Chat with Astrologer FAIL	api.whatsapp.com (domain only — SFSafariVC restriction)	SFSafariViewController	None verifiable	✗ All Missing
Financial Freedom FAIL	economictimes.indiatimes.com (domain only — SFSafariVC restriction)	SFSafariViewController	None verifiable	✗ All Missing

SFSafariViewController: Apple's iOS security model prevents the host app from reading the full URL — only the domain is accessible via `XCTest`. WhatsApp (`api.whatsapp.com`) also strips URL parameters on redirect, making UTM attribution via URL fundamentally impossible for that item.

FINDINGS SUMMARY

- **AI Masterclass** uses a non-standard `ag=toiapprails` parameter instead of the UTM standard. Traffic will not be correctly attributed in GA4 / Firebase / Amplitude as coming from the TOI app rail.
- **Chat with Astrologer** links to WhatsApp (`api.whatsapp.com`). Even if UTM params were appended, WhatsApp strips them on redirect — UTM attribution from this placement is fundamentally not possible via URL alone.
- **Financial Freedom** links to ET without any UTM parameters. TOI cannot measure the downstream value (traffic, revenue, conversions) of this rail placement on ET.

RECOMMENDED FIX — STANDARD UTM SCHEMA

- **AI Masterclass:** `?utm_source=toi_app&utm_medium=rail&utm_campaign=ai_masterclass_2026&utm_content=rail_ai_masterclass`
- **Chat with Astrologer:** Replace direct WhatsApp link with a tracked server-side redirect (e.g. Branch.io / Adjust) that logs the attribution event before forwarding to WhatsApp.
- **Financial Freedom:** `?utm_source=toi_app&utm_medium=rail&utm_campaign=financial_freedom_2026&utm_content=rail_financial_freedom`
- All outbound rail links should be validated via **Charles Proxy / mitmproxy** on-device or desktop browser inspection to confirm full UTM presence before release.

Result: 5/5 scenarios passed (100%) – PASS

#	TEMPLATE NAME	APP STATE	RESULT
1	Article Show	Killed	PASS
2	Live Blog	Background	PASS
3	Video	Foreground	PASS
4	Movie Review	Killed	PASS
5	Photo Story	Background	PASS

All 5 push notification templates were successfully received and displayed across different app states (Killed, Background, Foreground). Notification delivery and handling are working correctly.